Before/After Analysis — Capstone 64

Objective: determine whether AI assistance reduces EDA time while maintaining accuracy.

Current evidence

The analysis covered 1,000 student records and 10 columns. Data-quality checks reported zero missing values and zero duplicate rows. Descriptive summaries, cross-tabulations, correlation analysis, and visualizations were used to explore stress level, exam anxiety, academic performance change, screen time, sleep duration, and physical activity.

Comparison table

Metric	Manual/baseline	AI-assisted	Result
Time spent	30 minutes (estimated)	15 minutes (estimated)	50% estimated time saved
Data-quality checks	Manual review of dataset structure and quality	Python checks: missing values and duplicate rows	Both approaches support checking data quality; Python gives repeatable counts
Charts	Manual chart planning and creation	Charts generated with Python/Matplotlib/Seaborn	AI-assisted workflow makes chart creation faster to reproduce
Errors	No separately logged manual errors	One NameError appeared during notebook execution; this is a runtime issue, not a verified AI analysis error	No confirmed incorrect AI numerical claim identified in the provided results
Accuracy	Manual results used as conceptual baseline	Key counts checked against displayed Python outputs	Reported figures align with the shown outputs

Measurement summary

Baseline: 30 minutes (estimated) **AI-assisted:** 15 minutes (estimated)

Time saved: 50% (estimated) **Verified AI errors:** 0 confirmed

Conclusion

The AI-assisted approach is the better workflow for this EDA task: it helped organize the analysis, generate repeatable Python summaries, and produce visualizations efficiently. The displayed results were consistent across the reported counts and cross-tabulations, while the data-quality checks found no missing values or duplicate rows. Overall, the AI-assisted workflow offers a clearer and more efficient analysis process than a fully manual baseline.